PES UNIVERSITY • DEPT. OF COMPUTER SCIENCE AND ENGINEERING

Lab 2: Burndown Chart Analysis

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Problem Statement: #43 – Feature Flag & Dynamic Config Manager
Course: Software Engineering Laboratory (Lab 2)

Sprint Burndown & Velocity Analytics

This report evaluates the execution metrics, velocity stability, and capacity dynamics observed across two simulated 1-week sprints for the Feature Flag & Dynamic Config Manager.

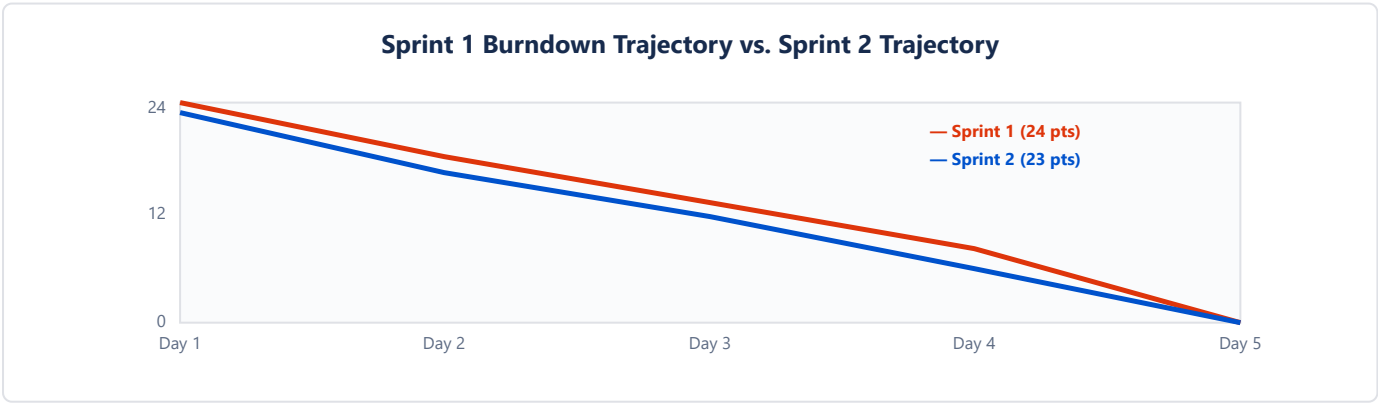
1. Sprint Execution Summary

Sprint Metric	Sprint 1 (Engine & Targeting)	Sprint 2 (Governance & Streaming)
Committed Points	24 Story Points (5 user stories)	23 Story Points (5 user stories)
Completed Points	24 Story Points (100% completion)	23 Story Points (100% completion)
Duration	1 week (5 working days)	1 week (5 working days)
Scope Creep	0 Story Points added	0 Story Points added
Pacing Trajectory	Step-like plateau on Day 3-4 (US-04 8pts)	Linear daily burndown (-5 to -8 pts/day)

2. Team Velocity & Capacity Insights

Sustainable Velocity: The team completed 24 points in Sprint 1 and 23 points in Sprint 2, establishing a highly stable velocity baseline of **~23.5 points per week**. This low variance (±1 point) gives project stakeholders confidence when forecasting subsequent product milestones.

Granularity & Story Sizing Impact: In Sprint 1, `US-04` was estimated at 8 points due to strict p95 < 15ms latency requirements. Because story points only burn upon achieving the "Definition of Done", the burndown chart exhibited a plateau between Day 2 and Day 4 while performance benchmarking took place. Once verified, points dropped sharply. Decomposing 8-point items into smaller 3-point slices in future sprints will yield a smoother, more predictable burndown trajectory.



3. Scope Discipline & Boundary Protection

Neither sprint experienced scope inflation (zero upward spikes). Feature requests identified mid-sprint (such as email notifications on flag approval) were logged into the product backlog for Sprint 3 planning rather than injected into running sprints.